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November 4, 2025, 3:50 PM

• Search:

CAS Lexicon Search:

Ring-closing metathesis - Preferred Concept

Search Tasks

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Returned Reference Results from CAS Lexicon (5,379)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (2,641)	 Reactions	View Results
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Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	
Publication Year:	2015 to 2025	

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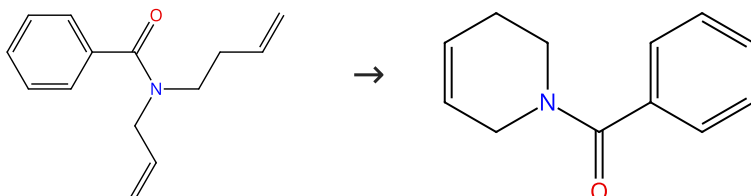
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (5)

31-470-CAS-5224012

Steps: 1 Yield: 100%

A medium fluororous Grubbs-Hoveyda 2nd generation catalyst for phase transfer catalysis of ring closing metathesis reactions

1.1 Catalysts: 1661836-68-5

Solvents: Dimethylformamide; 1 h, rt

1.2 Solvents: Dimethylformamide, Water; 5 min, 5 °C

Experimental Protocols

By: Kobayashi, Yuki; et al

Tetrahedron Letters (2015), 56(11), 1363-1366.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-250-CAS-18707208

Steps: 1 Yield: 100%

Diastereoselective Synthesis of an Advanced Intermediate of Thapsigargin and Other 6,12-Guaianolides Using a RCEYM Strategy

1.1 Reagents: Ethylene

Catalysts: Grubbs second generation catalyst

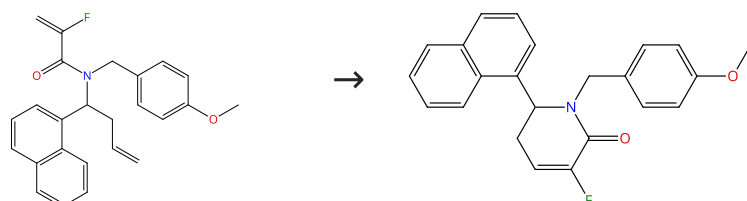
Solvents: Toluene; rt → 80 °C; 15 h, 80 °C

By: Jouanneau, Morgan; et al

Organic Letters (2018), 20(8), 2176-2180.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-1305810

Steps: 1 Yield: 100%

Short and efficient synthesis of fluorinated δ -lactams

1.1 Catalysts: Grubbs second generation catalyst

Solvents: Toluene; 1 - 4 h, rt → 100 °C

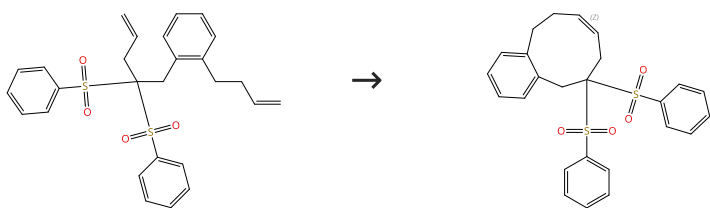
By: Cogswell, Thomas J.; et al

Organic & Biomolecular Chemistry (2015), 13(3), 717-728.

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

31-470-CAS-16288665

Steps: 1 Yield: 100%

Synthesis and Stereochemical Analysis of Planar-Chiral (E)-4-[7]Orthocyclophene

1.1 Catalysts: Grubbs' catalyst

Solvents: Dichloromethane; 8 h, reflux; 22 h, reflux

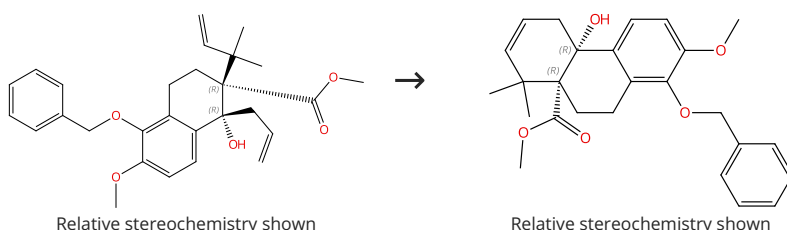
By: Igawa, Kazunobu; et al

Experimental Protocols

Journal of Organic Chemistry (2016), 81(23), 11587-11593.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Relative stereochemistry shown

31-470-CAS-22426881

Steps: 1 Yield: 100%

Formal Total Synthesis of Atropurpuran

1.1 Reagents: Grubbs second generation catalyst

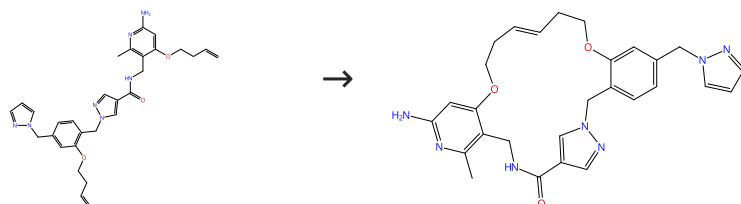
Solvents: Dichloromethane

By: Suzuki, Takahiro; et al

Journal of Organic Chemistry (2020), 85(15), 10125-10135.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-16703349

Steps: 1 Yield: 100%

Structure-Guided Design of Novel, Potent, and Selective Macrocyclic Plasma Kallikrein Inhibitors

1.1 Catalysts: Grubbs second generation catalyst

Solvents: 1,2-Dichloroethane; 12 - 48 h, 70 °C

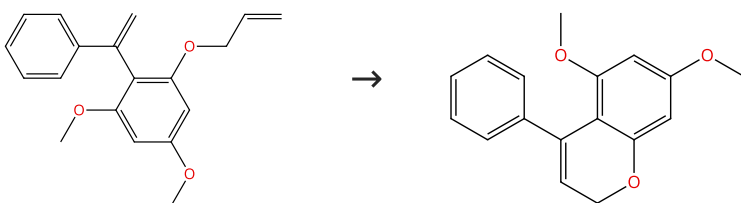
By: Li, Zhe; et al

Experimental Protocols

ACS Medicinal Chemistry Letters (2017), 8(2), 185-190.

Scheme 7 (1 Reaction)

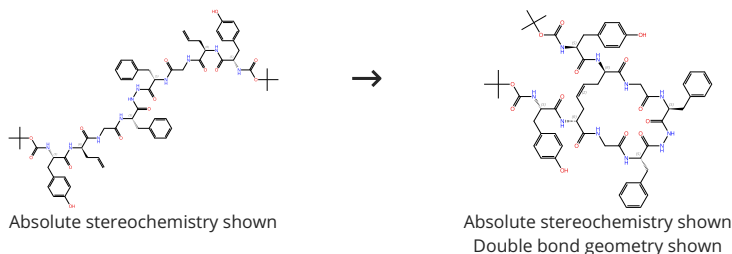
Steps: 1 Yield: 100%



31-614-CAS-33444507	Steps: 1 Yield: 100%	Ring-closing metathesis in flavonoid synthesis, part 2: neoflav-3-enes
1.1 Catalysts: Grubbs second generation catalyst Solvents: Dichloromethane; rt → reflux; overnight, reflux		By: Pieterse, Tanya; et al
Experimental Protocols		ARKIVOC (Gainesville, FL, United States) (2022), (5), 217-231.

Scheme 8 (1 Reaction)

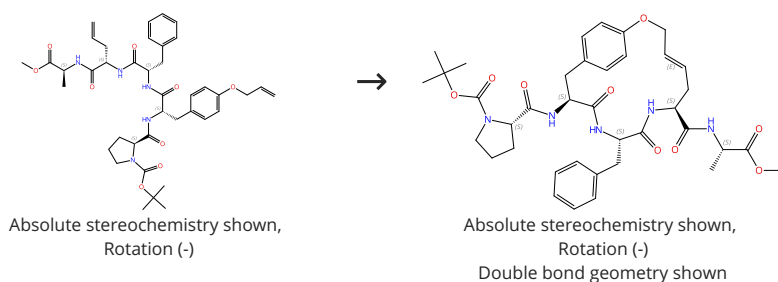
Steps: 1 Yield: 100%



31-614-CAS-35394235	Steps: 1 Yield: 100%	Novel Cyclic Biphalin Analogues by Ruthenium-Catalyzed Ring Closing Metathesis: in Vivo and in Vitro Biological Profile
1.1 Catalysts: Grubbs second generation catalyst Solvents: 1,2-Dichloroethane; 78 h, 80 °C		By: Stefanucci, Azzurra; et al
Experimental Protocols		ACS Medicinal Chemistry Letters (2019), 10(4), 450-456.

Scheme 9 (1 Reaction)

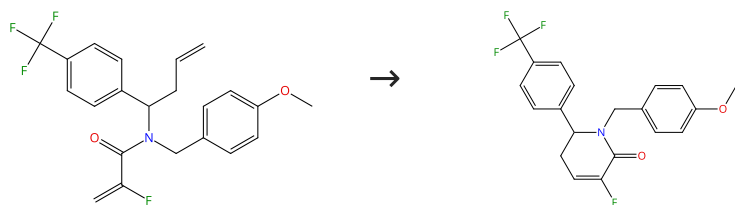
Steps: 1 Yield: 100%



31-470-CAS-22713731	Steps: 1 Yield: 100%	Design and synthesis of β -strand-fixed peptides inhibiting aggregation of amyloid β -protein
1.1 Catalysts: (SP-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy- κ O)phenyl]methylene- κ C]ruthenium Solvents: Dichloromethane; rt; 23 h, reflux		By: Tanaka, Fumiya; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2020), 28(18), 115676.

Scheme 10 (1 Reaction)

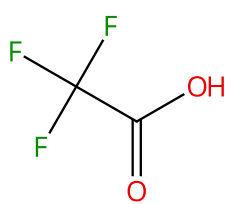
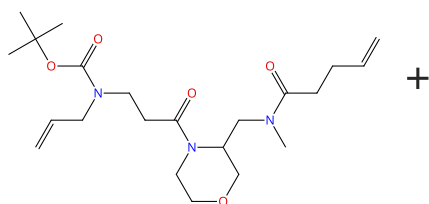
Steps: 1 Yield: 100%



31-470-CAS-1030523	Steps: 1 Yield: 100%	Short and efficient synthesis of fluorinated δ -lactams
1.1 Catalysts: Grubbs second generation catalyst Solvents: Toluene; 1 - 4 h, rt → 100 °C		By: Cogswell, Thomas J.; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2015), 13(3), 717-728.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Multi-component
structure image
available in CAS
SciFinder

Suppliers (192)

31-353-CAS-23737028

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** (SP-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy-κO)phenyl]methylene-κC]ruthenium
Solvents: Dichloromethane; 24 h, rt
- 1.2 **Reagents:** Triisopropylsilane
Solvents: Water; 6 h, rt

Experimental Protocols

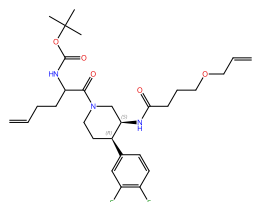
A Diversity-Oriented Approach to Large Libraries of Artificial Macrocycles

By: Kharchenko, Serhii H.; et al

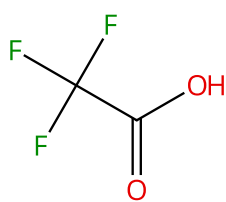
European Journal of Organic Chemistry (2021), 2021(17), 2313-2330.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown



Multi-component
structure image
available in CAS
SciFinder

Suppliers (192)

31-353-CAS-23738849

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** (SP-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy-κO)phenyl]methylene-κC]ruthenium
Solvents: Dichloromethane; 24 h, rt
- 1.2 **Reagents:** Triisopropylsilane
Solvents: Water; 6 h, rt

Experimental Protocols

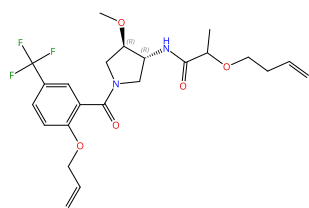
A Diversity-Oriented Approach to Large Libraries of Artificial Macrocycles

By: Kharchenko, Serhii H.; et al

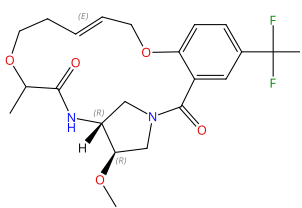
European Journal of Organic Chemistry (2021), 2021(17), 2313-2330.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Relative stereochemistry shown
Double bond geometry shown

Supplier (1)

31-470-CAS-23738741

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** (SP-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy-κO)phenyl]methylene-κC]ruthenium
Solvents: Dichloromethane; 24 h, rt

Experimental Protocols

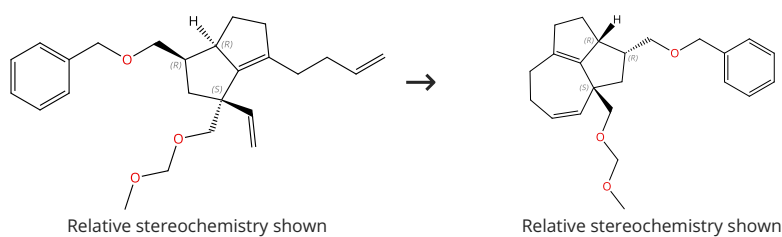
A Diversity-Oriented Approach to Large Libraries of Artificial Macrocycles

By: Kharchenko, Serhii H.; et al

European Journal of Organic Chemistry (2021), 2021(17), 2313-2330.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-18972809

Steps: 1 Yield: 100%

Synthesis of the [7-5-5] tricyclic core of Daphniphyllum alkaloids

1.1 **Catalysts:** Grubbs second generation catalyst
Solvents: 1,2-Dichloroethane; 25 h, 50 °C

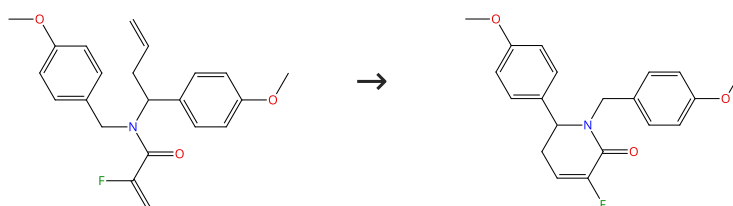
By: Kitabayashi, Yusuke; et al

Experimental Protocols

Organic & Biomolecular Chemistry (2018), 16(19), 3556-3559.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-12300323

Steps: 1 Yield: 100%

Short and efficient synthesis of fluorinated δ -lactams

1.1 **Catalysts:** Grubbs second generation catalyst
Solvents: Toluene; 1 - 4 h, rt $\rightarrow$ 100 °C

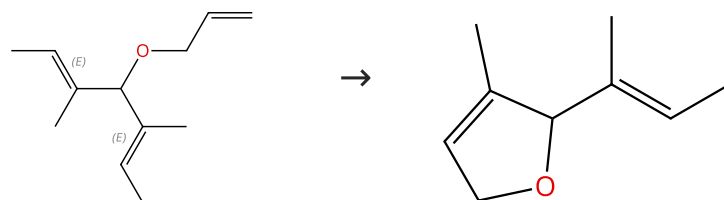
By: Cogswell, Thomas J.; et al

Experimental Protocols

Organic & Biomolecular Chemistry (2015), 13(3), 717-728.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Suppliers (2)

Suppliers (5)

31-470-CAS-18278772

Steps: 1 Yield: 100%

Amino acids as chiral anionic ligands for ruthenium based asymmetric olefin metathesis

1.1 **Catalysts:** 1660159-48-7
Solvents: Tetrahydrofuran; 2 h, 37 °C

By: Ivry, Elisa; et al

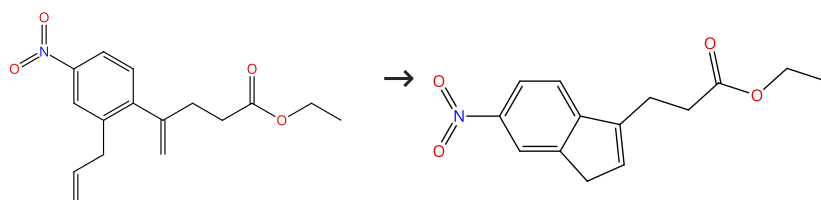
1.2 **Reagents:** Ethyl vinyl ether

Experimental Protocols

Chemical Communications (Cambridge, United Kingdom) (2015), 51(18), 3870-3873.

Scheme 17 (1 Reaction)

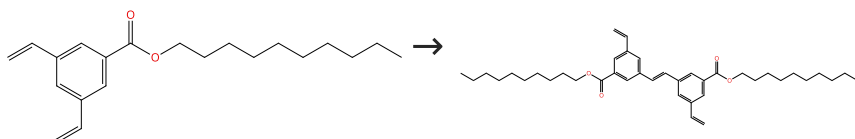
Steps: 1 Yield: 100%



31-470-CAS-16954182 Steps: 1 Yield: 100% 1.1 Catalysts: (SP-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy-κO)-5-nitrophenyl]methylene-κC]ruthenium Solvents: Dichloromethane; 1 h, 40 °C	Synthesis of Selectively Substituted or Deuterated Indenes via Sequential Pd and Ru Catalysis By: Jana, Anupam; et al Journal of Organic Chemistry (2017), 82(8), 4226-4234.
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Scheme 18 (1 Reaction)

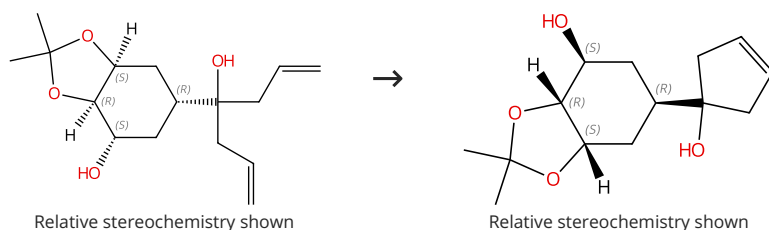
Steps: 1 Yield: 100%



31-470-CAS-19205691 Steps: 1 Yield: 100% 1.1 18 h, 35 °C	Surface-Confined Dynamic Covalent System Driven by Olefin Metathesis By: Liu, Chunhua; et al Angewandte Chemie, International Edition (2018), 57(7), 1869-1873.
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Scheme 19 (1 Reaction)

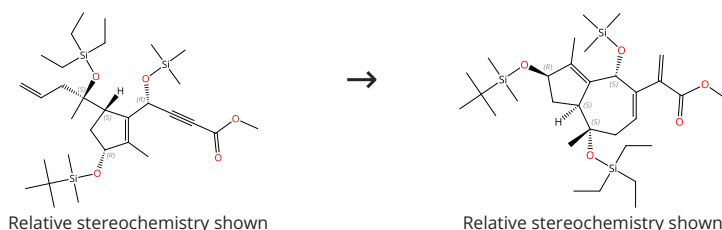
Steps: 1 Yield: 100%



31-470-CAS-7572720 Steps: 1 Yield: 100% 1.1 Catalysts: Grubbs second generation catalyst Solvents: Toluene; 1 h, 60 °C	Conversion of Ester Moieties to 4-Bromophenyl Groups via Electrocyclic Reaction of Dibromocyclopropanes By: Ueda, Kyosuke; et al Organic Letters (2015), 17(13), 3191-3193.
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Scheme 20 (1 Reaction)

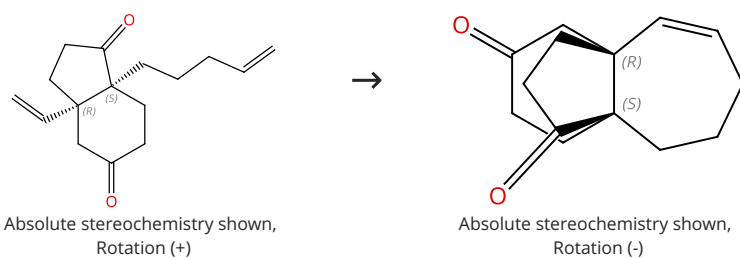
Steps: 1 Yield: 100%



31-250-CAS-18707207 Steps: 1 Yield: 100% 1.1 Reagents: Ethylene Catalysts: Grubbs second generation catalyst Solvents: Toluene; rt → 80 °C; 15 h, 80 °C	Diastereoselective Synthesis of an Advanced Intermediate of Thapsigargin and Other 6,12-Guaianolides Using a RCEYM Strategy By: Jouanneau, Morgan; et al Organic Letters (2018), 20(8), 2176-2180.
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Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-17111034

Steps: 1 Yield: 100%

Asymmetric Synthesis of Carbocyclic Propellanes

- 1.1 **Catalysts:** Quinone, (*SP*-5-41)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy-κO)-5-[(2,2,2-trifluoroacetyl)amino]phenyl)methylene-κC] ruthenium
- Solvents:** 1,2-Dichloroethane; 2.5 h, 85 °C

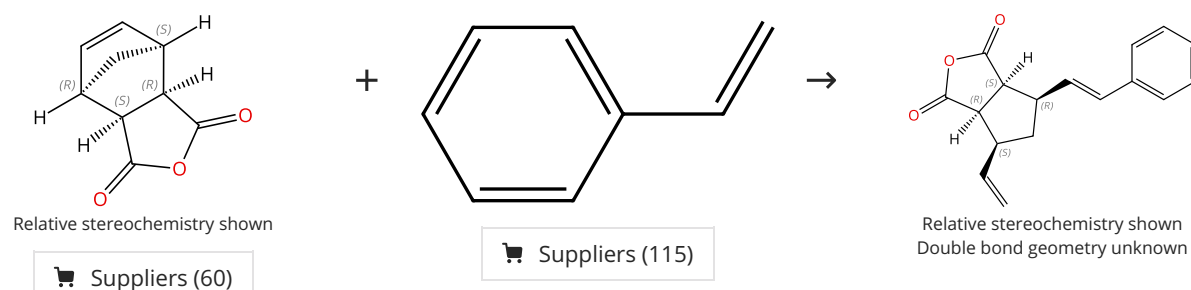
By: Schneider, Lisa M.; et al

Organic Letters (2017), 19(9), 2310-2313.

Experimental Protocols

Scheme 22 (4 Reactions)

Steps: 1 Yield: 100%



31-470-CAS-12200534

Steps: 1 Yield: 100%

Amino acids as chiral anionic ligands for ruthenium based asymmetric olefin metathesis

- 1.1 **Catalysts:** 1660159-46-5
- Solvents:** Tetrahydrofuran; 2 h, 37 °C

By: Ivry, Elisa; et al

- 1.2 **Reagents:** Ethyl vinyl ether

Chemical Communications (Cambridge, United Kingdom) (2015), 51(18), 3870-3873.

Experimental Protocols

31-470-CAS-5517232

Steps: 1 Yield: 100%

Amino acids as chiral anionic ligands for ruthenium based asymmetric olefin metathesis

- 1.1 **Catalysts:** 1660159-47-6
- Solvents:** Tetrahydrofuran; 2 h, 37 °C

By: Ivry, Elisa; et al

- 1.2 **Reagents:** Ethyl vinyl ether

Chemical Communications (Cambridge, United Kingdom) (2015), 51(18), 3870-3873.

Experimental Protocols

31-470-CAS-5186195

Steps: 1 Yield: 100%

Amino acids as chiral anionic ligands for ruthenium based asymmetric olefin metathesis

- 1.1 **Catalysts:** 1660159-49-8
- Solvents:** Tetrahydrofuran; 2 h, 37 °C

By: Ivry, Elisa; et al

- 1.2 **Reagents:** Ethyl vinyl ether

Chemical Communications (Cambridge, United Kingdom) (2015), 51(18), 3870-3873.

Experimental Protocols

31-470-CAS-946815

Steps: 1 Yield: 100%

Amino acids as chiral anionic ligands for ruthenium based asymmetric olefin metathesis

- 1.1 **Catalysts:** 1669427-16-0
- Solvents:** Tetrahydrofuran; 2 h, 37 °C

By: Ivry, Elisa; et al

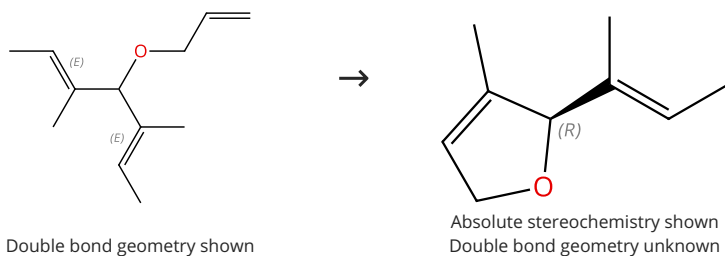
- 1.2 **Reagents:** Ethyl vinyl ether

Chemical Communications (Cambridge, United Kingdom) (2015), 51(18), 3870-3873.

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

31-470-CAS-18051076

Steps: 1 Yield: 100%

Amino acids as chiral anionic ligands for ruthenium based asymmetric olefin metathesis1.1 **Catalysts:** 1660159-47-6**Solvents:** Tetrahydrofuran; 2 h, 37 °C1.2 **Reagents:** Ethyl vinyl ether

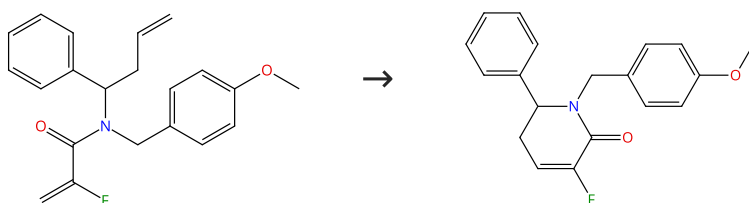
Experimental Protocols

By: Ivry, Elisa; et al

Chemical Communications (Cambridge, United Kingdom) (2015), 51(18), 3870-3873.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-11512613

Steps: 1 Yield: 100%

Short and efficient synthesis of fluorinated δ -lactams1.1 **Catalysts:** Grubbs second generation catalyst**Solvents:** Toluene; 1 - 4 h, rt $\rightarrow$ 100 °C

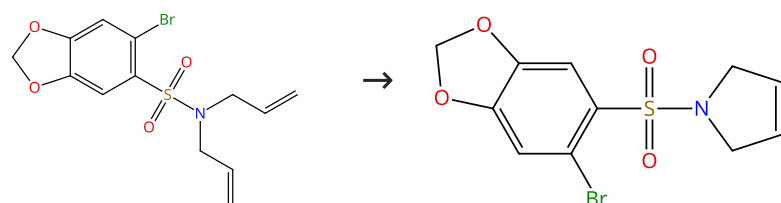
Experimental Protocols

By: Cogswell, Thomas J.; et al

Organic & Biomolecular Chemistry (2015), 13(3), 717-728.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

31-470-CAS-15841596

Steps: 1 Yield: 100%

A ring closing metathesis-manganese dioxide oxidation sequence for the synthesis of substituted pyrroles1.1 **Catalysts:** Grubbs' catalyst**Solvents:** Dichloromethane; 15 h, 1 °C

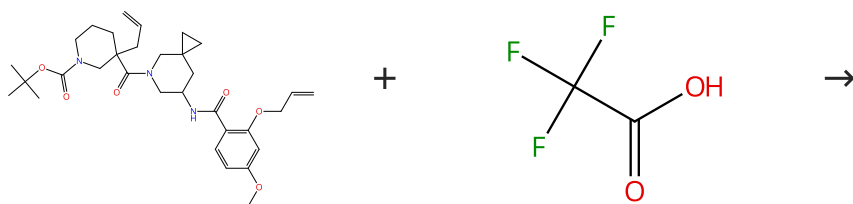
Experimental Protocols

By: Keeley, Aaron; et al

Tetrahedron (2016), 72(20), 2552-2559.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (192)

Multi-component
structure image
available in CAS
SciFinder

31-353-CAS-23734446

Steps: 1 Yield: 100%

A Diversity-Oriented Approach to Large Libraries of Artificial Macrocycles

By: Kharchenko, Serhii H.; et al

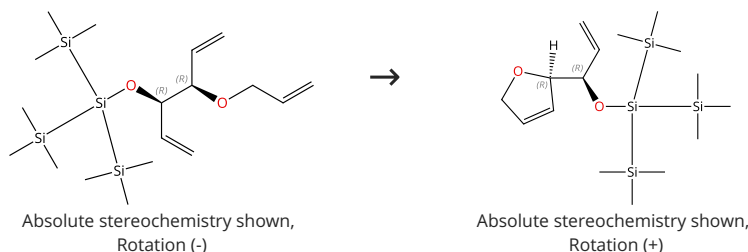
European Journal of Organic Chemistry (2021), 2021(17), 2313-2330.

- 1.1 **Catalysts:** (*SP*-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy- κ O)phenyl]methylene- κ C]ruthenium
Solvents: Dichloromethane; 24 h, rt
- 1.2 **Reagents:** Triisopropylsilane
Solvents: Water; 6 h, rt

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-5966327

Steps: 1 Yield: 100%

Regioselective Chromatic Orthogonality with Light-Activated Metathesis Catalysts

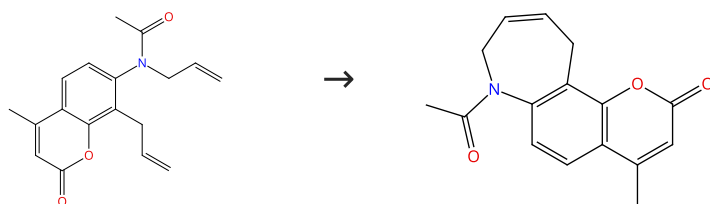
By: Levin, Efrat; et al

Angewandte Chemie, International Edition (2015), 54(42), 12384-12388.

- 1.1 **Catalysts:** (*SP*-5-41)-Dichloro[[2-(1-methylethoxy- κ O)phenyl]methylene- κ C](tricyclohexylphosphine)ruthenium
Solvents: Dichloromethane; rt

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

31-470-CAS-20249461

Steps: 1 Yield: 100%

Ring-closing-metathesis-based synthesis of annelated coumarins from 8-allylcoumarins

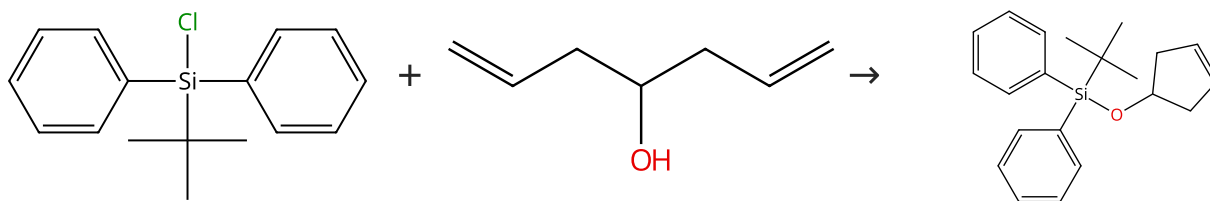
By: Schultze, Christiane; et al

Beilstein Journal of Organic Chemistry (2018), 14, 2991-2998.

- 1.1 **Catalysts:** Grubbs second generation catalyst
Solvents: Toluene; 1 h, rt $\rightarrow$ 90 °C

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (51)

Suppliers (31)

31-008-CAS-8408006

Steps: 1 Yield: 100%

A general and scalable synthesis of aeruginosin marine natural products based on two strategic C(sp³)-H activation reactions

By: Dailier, David; et al

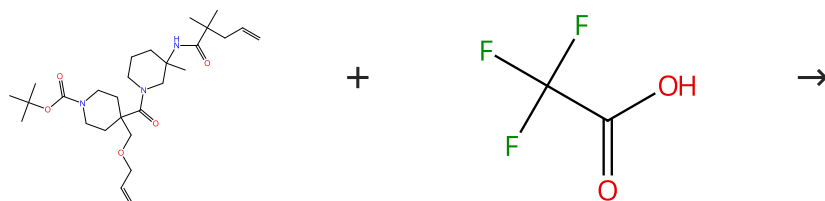
Angewandte Chemie, International Edition (2015), 54(16), 4919-4922.

- 1.1 **Reagents:** Imidazole
Solvents: Dichloromethane; 0 °C; 16 h, rt
- 1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; rt
- 1.3 **Catalysts:** Grubbs' catalyst
Solvents: Dichloromethane; 20 min, rt

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (192)

Multi-component
structure image
available in CAS
SciFinder

31-353-CAS-23738565

Steps: 1 Yield: 100%

A Diversity-Oriented Approach to Large Libraries of Artificial Macrocycles

By: Kharchenko, Serhii H.; et al

European Journal of Organic Chemistry (2021), 2021(17), 2313-2330.

- 1.1 **Catalysts:** (*SP*-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy-κ*O*)phenyl]methylene-κ*C*]ruthenium
Solvents: Dichloromethane; 24 h, rt
- 1.2 **Reagents:** Triisopropylsilane
Solvents: Water; 6 h, rt

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



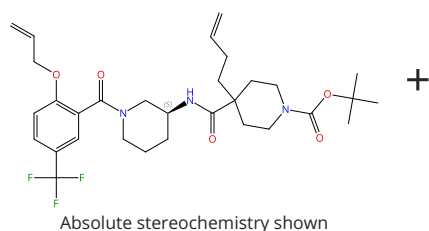
Suppliers (16)

Suppliers (7)

31-470-CAS-6054281 Steps: 1 Yield: 100% 1.1 Reagents: 1,4-Diacetoxy-2-butene Catalysts: (<i>SP</i> -5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy- κO)-5-nitrophenyl]methylene- κC]ruthenium Solvents: Toluene; 1 h, 70 °C 1.2 Reagents: 1,4-Bis(3-isocyanopropyl)piperazine Solvents: Methanol; 30 min, rt Experimental Protocols	High-Performance Isocyanide Scavengers for Use in Low-Waste Purification of Olefin Metathesis Products By: Szczepaniak, Grzegorz; et al ChemSusChem (2015), 8(24), 4139-4148.
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Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



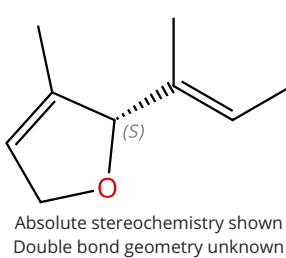
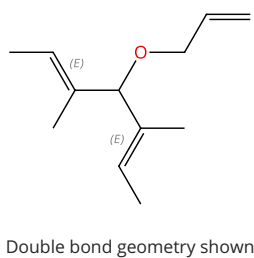
Suppliers (192)

Multi-component
structure image
available in CAS
SciFinder

31-353-CAS-23738815 Steps: 1 Yield: 100% 1.1 Catalysts: (<i>SP</i> -5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy- κO)phenyl]methylene- κC]ruthenium Solvents: Dichloromethane; 24 h, rt 1.2 Reagents: Triisopropylsilane Solvents: Water; 6 h, rt Experimental Protocols	A Diversity-Oriented Approach to Large Libraries of Artificial Macrocycles By: Kharchenko, Serhii H.; et al European Journal of Organic Chemistry (2021), 2021(17), 2313-2330.
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Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%

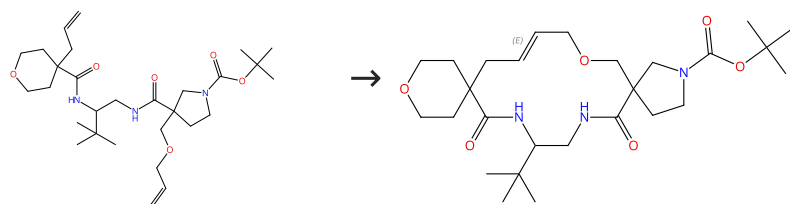


Suppliers (2)

31-470-CAS-18070318 Steps: 1 Yield: 100% 1.1 Catalysts: 1669427-16-0 Solvents: Tetrahydrofuran; 2 h, 37 °C 1.2 Reagents: Ethyl vinyl ether Experimental Protocols	Amino acids as chiral anionic ligands for ruthenium based asymmetric olefin metathesis By: Ivry, Elisa; et al Chemical Communications (Cambridge, United Kingdom) (2015), 51(18), 3870-3873.
---	---

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Double bond geometry shown

31-470-CAS-23737781

Steps: 1 Yield: 100%

A Diversity-Oriented Approach to Large Libraries of Artificial Macrocycles

By: Kharchenko, Serhii H.; et al

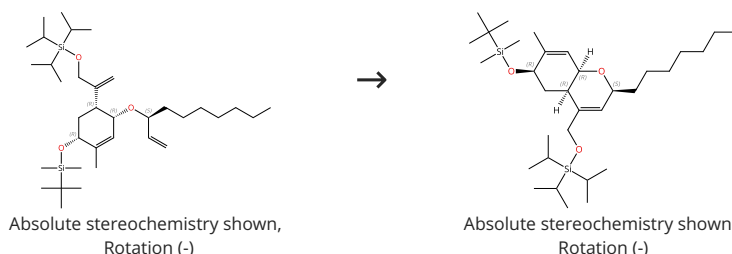
European Journal of Organic Chemistry (2021), 2021(17), 2313-2330.

1.1 **Catalysts:** (*SP*-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy- κ O)phenyl]methylene- κ C]ruthenium
Solvents: Dichloromethane; 24 h, rt

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (-)

31-614-CAS-36683484

Steps: 1 Yield: 100%

Synthesis of Stereoisomeric Simplified Analogs of Alotak etals toward the Elucidation of the Structural Requirements of Protein Kinase C Isozyme-Selective Binding

By: Okuda, Sogen; et al

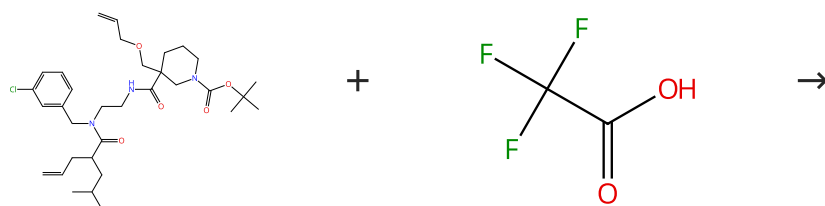
Organic Letters (2023), 25(5), 805-809.

1.1 **Catalysts:** Grubbs second generation catalyst
Solvents: 1,2-Dichloroethane; 2 h, rt; 3 h, 50 °C

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (192)

Multi-component
structure image
available in CAS
SciFinder

31-353-CAS-23732753

Steps: 1 Yield: 100%

A Diversity-Oriented Approach to Large Libraries of Artificial Macrocycles

By: Kharchenko, Serhii H.; et al

European Journal of Organic Chemistry (2021), 2021(17), 2313-2330.

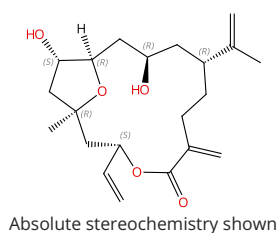
1.1 **Catalysts:** (*SP*-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy- κ O)phenyl]methylene- κ C]ruthenium
Solvents: Dichloromethane; 24 h, rt

1.2 **Reagents:** Triisopropylsilane
Solvents: Water; 6 h, rt

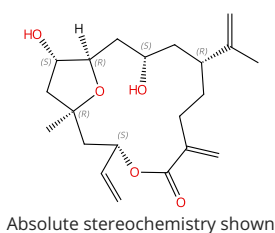
Experimental Protocols

Scheme 37 (1 Reaction)

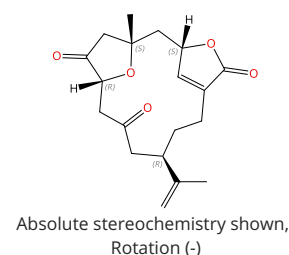
Steps: 1 Yield: 100%



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31-614-CAS-44744356

Steps: 1 Yield: 100%

Total synthesis and structure-antifouling activity relationship of scabrolide F

By: Takamura, Hiroyoshi; et al

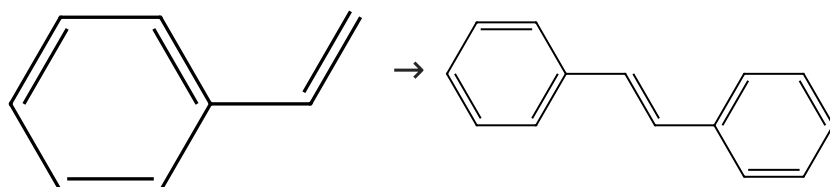
Organic & Biomolecular Chemistry (2024), 22(28), 5739-5747.

- 1.1 **Reagents:** Dess-Martin periodinane
Solvents: Dichloromethane; 0 °C; 3 h, rt
- 1.2 **Reagents:** Sodium thiosulfate
Solvents: Water
- 1.3 **Catalysts:** (*SP*-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy- κ O)phenyl]methylene- κ C]ruthenium
Solvents: Toluene; rt; 2 h, 80 °C

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (115)

Suppliers (11)

31-470-CAS-23044203

Steps: 1 Yield: 100%

Influence of the N→Ru Coordinate Bond Length on the Activity of New Types of Hoveyda-Grubbs Olefin Metathesis Catalysts Containing a Six-Membered Chelate Ring Possessing a Ruthenium-Nitrogen Bond

By: Kumandin, Pavel A.; et al

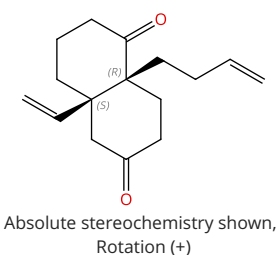
Organometallics (2020), 39(24), 4599-4607.

- 1.1 **Catalysts:** (*SP*-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-[(4-morpholinyl- κ N⁴)methyl]phenyl]methylene- κ C]ruthenium
Solvents: Dichloromethane; 1 h, reflux

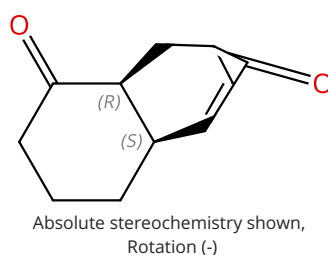
Experimental Protocols

Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



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31-470-CAS-17157703

Steps: 1 Yield: 100%

Asymmetric Synthesis of Carbocyclic Propellanes

By: Schneider, Lisa M.; et al

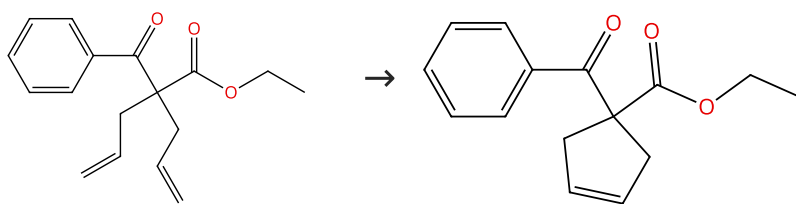
Organic Letters (2017), 19(9), 2310-2313.

- 1.1 **Catalysts:** (*SP*-5-41)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy- κ O)-5-[(2,2,2-trifluoroacetyl)amino]phenyl]methylene- κ C]ruthenium
Solvents: Toluene; 1 h, 110 °C

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

Suppliers (4)

31-470-CAS-2409646

Steps: 1 Yield: 100%

High-Performance Isocyanide Scavengers for Use in Low-Waste Purification of Olefin Metathesis Products

By: Szczepaniak, Grzegorz; et al

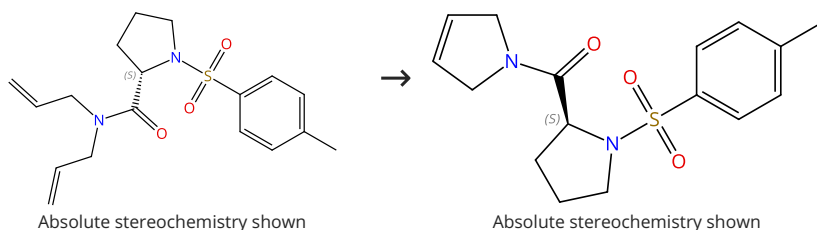
ChemSusChem (2015), 8(24), 4139-4148.

- 1.1 **Reagents:** 1,4-Diacetoxy-2-butene
Catalysts: (*SP*-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy-κ*O*)-5-nitrophenyl]methylene-κ*C*]ruthenium
Solvents: Toluene; 1 h, 70 °C
- 1.2 **Reagents:** 1,4-Bis(3-isocyanopropyl)piperazine
Solvents: Methanol; 30 min, rt

Experimental Protocols

Scheme 41 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-14153610

Steps: 1 Yield: 100%

High-Performance Isocyanide Scavengers for Use in Low-Waste Purification of Olefin Metathesis Products

By: Szczepaniak, Grzegorz; et al

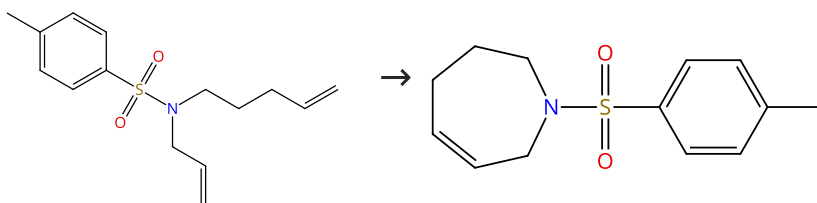
ChemSusChem (2015), 8(24), 4139-4148.

- 1.1 **Catalysts:** Ruthenium(1+), dichloro[[2-(1-methylethoxy-κ*O*)phenyl]methylene-κ*C*][4-[(trimethylammonio)methyl]-1,3-bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]-, iodide (1:1), (*S* *P*-5-41)-
Solvents: Ethyl acetate; 1 h, 70 °C
- 1.2 **Reagents:** 1,4-Bis(3-isocyanopropyl)piperazine; 30 min, 70 °C

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (5)

31-470-CAS-18495663

Steps: 1 Yield: 100%

Light- and Thermal-Activated Olefin Metathesis of Hindered Substrates

By: Ivry, Elisa; et al

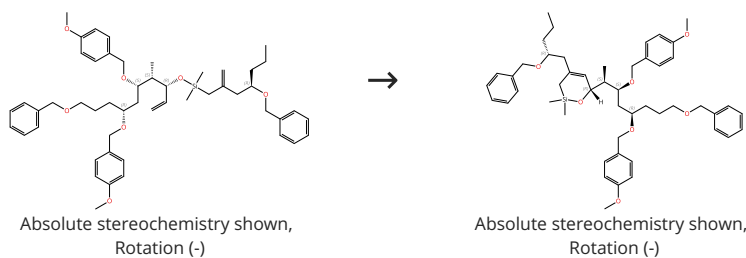
Organometallics (2018), 37(2), 176-181.

- 1.1 **Catalysts:** Ruthenium, [1,3-bis[2,6-bis(1-methylethyl)phenyl]-2-imidazolidinylidene]dichloro[[2-[(trifluoromethyl)thio-κ*S*]phenyl]methylene-κ*C*]-, stereoisomer
Solvents: Toluene; < 15 min, 80 °C

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-19638673

Steps: 1 Yield: 100%

Novel Synthesis of Trisubstituted Olefins for the Preparation of the C16-C30 Fragment of Dolabelide C

By: Tiniakos, Alexander F.; et al

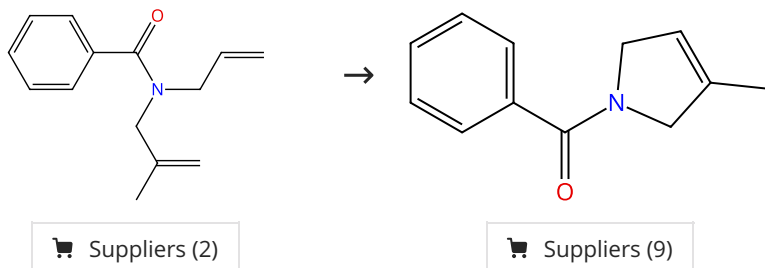
Organic Letters (2019), 21(3), 589-592.

1.1 **Catalysts:** (*SP*-5-41)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro[[2-(1-methylethoxy-κ*O*)-5-[(2,2,2-trifluoroacetyl)amino]phenyl]methylene-κ*C*]ruthenium
Solvents: Toluene; 1 d, 70 °C; 24 h, 70 °C

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 100%



31-470-CAS-2745255

Steps: 1 Yield: 100%

A medium fluoruous Grubbs-Hoveyda 2nd generation catalyst for phase transfer catalysis of ring closing metathesis reactions

By: Kobayashi, Yuki; et al

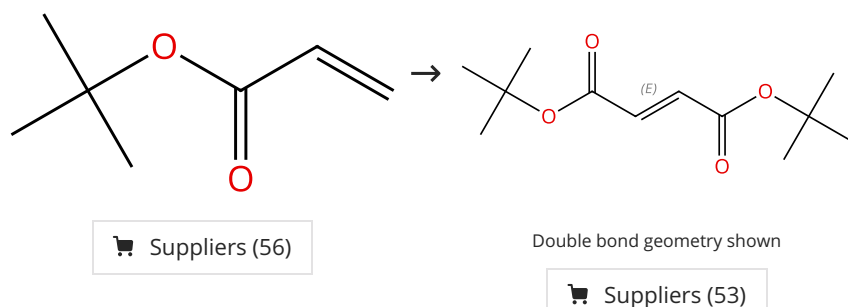
Tetrahedron Letters (2015), 56(11), 1363-1366.

1.1 **Catalysts:** 1661836-68-5
Solvents: Dimethylformamide; 4 h, rt
 1.2 **Solvents:** Dimethylformamide, Water; 5 min, 5 °C

Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-39097095

Steps: 1 Yield: 100%

Ruthenium-catalyzed Olefin Metathesis in Water using Thermo-responsive Diblock Copolymer Micelles

By: Suzuki, Noriyuki; et al

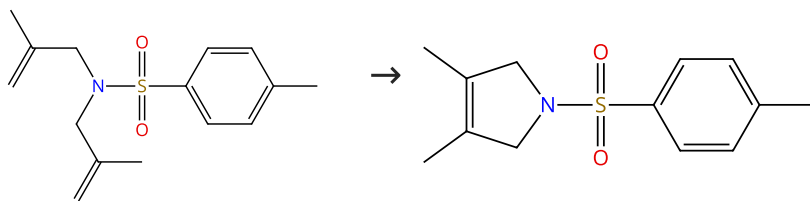
Current Organic Chemistry (2023), 27(15), 1347-1356.

1.1 **Reagents:** 1-Propanesulfonic acid, 2-methyl-2-[(1-oxo-2-propen-1-yl)amino]-, sodium salt (1:1), polymer with *N*-(1-methylethyl)-2-propenamide, diblock
Catalysts: (*SP*-5-41)-Dichloro[[2-(1-methylethoxy-κ*O*)phenyl]methylene-κ*C*](tricyclohexylphosphine)ruthenium
Solvents: Water; 12 h, 50 °C

Experimental Protocols

Scheme 46 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

Suppliers (2)

31-470-CAS-16209487

Steps: 1 Yield: 100%

Highly Activated Second-Generation Grubbs-Hoveyda Catalyst Driven by Intramolecular Steric Strain

1.1 Catalysts: 1972599-81-7

Solvents: Octafluorotoluene; 10 h, 60 °C

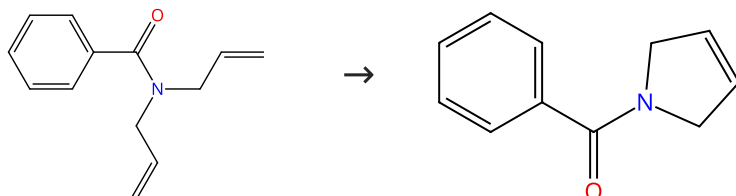
By: Kobayashi, Yuki; et al

Experimental Protocols

Synlett (2016), 27(16), 2352-2356.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (13)

Suppliers (25)

31-470-CAS-1147528

Steps: 1 Yield: 100%

A medium fluororous Grubbs-Hoveyda 2nd generation catalyst for phase transfer catalysis of ring closing metathesis reactions

1.1 Catalysts: 1661836-68-5

Solvents: Dimethylformamide; 1 h, rt

By: Kobayashi, Yuki; et al

1.2 Solvents: Dimethylformamide, Water; 5 min, 5 °C

Experimental Protocols

Tetrahedron Letters (2015), 56(11), 1363-1366.